# Worksheet 7.1 — Average & Instantaneous Rate of Change

Grade 12 Advanced Functions (MHF4U) · Unit 7: Rates of Change and Combining Functions

## The Big Ideas

The average rate of change over $[a, b]$ is the secant slope $\dfrac{f(b)-f(a)}{b-a}$; the instantaneous rate at a point is the tangent slope, estimated over a tiny interval.

- Average rate $= \dfrac{f(b)-f(a)}{b-a}$ (secant slope).
- Instantaneous rate = tangent slope at a point.
- Estimate it with a small interval like $[a, a+0.1]$.

## Core Concepts

### Average rate of change

The **average rate of change** over an interval is the slope of the secant line, $\dfrac{f(b)-f(a)}{b-a}$.

### Instantaneous rate

The **instantaneous rate** at a point is the slope of the tangent line, approached by shrinking the interval toward that point.

### Estimating it

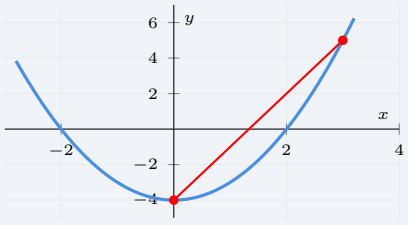
Use average rates over smaller and smaller intervals around a point to estimate the instantaneous rate there.

## Worked Examples

### Example 1 — Average rate

Average rate of change of $f(x) = x^2 - 4$ on $[0, 3]$.

***Solution.*** $\dfrac{f(3)-f(0)}{3-0} = \dfrac{5-(-4)}{3} = 3$. The secant slope approaches the tangent as the interval shrinks:




### Example 2 — Average rate

Average rate of change of $f(x) = 3x^2$ on $[1, 3]$.

***Solution.*** $\dfrac{27-3}{2} = 12$.

### Example 3 — Estimate instantaneous

Estimate the instantaneous rate of $f(x) = x^2$ at $x = 4$ using $[4, 4.1]$.

***Solution.*** $\dfrac{4.1^2 - 4^2}{0.1} = \dfrac{16.81 - 16}{0.1} = 8.1$ (true value 8).

### Example 4 — Linear rate

Average rate of change of $f(x) = 4x - 3$ on $[1, 6]$.

***Solution.*** $\dfrac{21 - 1}{5} = 4$ constant for a line.

### Example 5 — Interpret

A car's position is $d(t)$. What is $\dfrac{d(9) - d(3)}{9 - 3}$?

***Solution.*** The average speed over $[3, 9]$.

### Example 6 — Average rate

Average rate of $f(x) = x^2$ on $[3, 7]$.

***Solution.*** $\dfrac{49 - 9}{4} = 10$.

### Example 7 — Cubic

Average rate of $f(x) = x^3$ on $[2, 3]$.

***Solution.*** $\dfrac{27 - 8}{1} = 19$.

### Example 8 — Estimate instantaneous

Estimate the instantaneous rate of $f(x) = x^3$ at $x = 1$ using $[1, 1.1]$.

***Solution.*** $\dfrac{1.331 - 1}{0.1} = 3.31 \approx 3$.

### Example 9 — Tangent meaning

What does a tangent line's slope measure?

***Solution.*** The instantaneous rate of change at that point.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

Average rate of $f(x) = x^2 - 4x$ on $[1, 5]$?

### Question 2

Average rate of $f(x) = 2x^3$ on $[1, 2]$?

### Question 3

Estimate the instantaneous rate of $f(x) = x^2$ at $x = 5$ using $[5, 5.1]$.

### Question 4

Average rate of $f(x) = 7x + 1$ on $[2, 5]$?

### Question 5

What does the slope of a tangent line measure?

## Question 6

Average rate of $f(x) = x^2$ on $[2, 8]$?

## Question 7

Average rate of $f(x) = x^2 + 1$ on $[0, 3]$?

## Question 8

Average rate of $f(x) = 3x$ on $[2, 6]$?

## Question 9

Estimate the instantaneous rate of $f(x) = x^2$ at $x = 6$ using $[6, 6.1]$.

## Question 10

Average rate of $f(x) = x^3$ on $[0, 2]$?

## Question 11

Does a line have a changing instantaneous rate?

## Question 12

Average rate of $f(x) = -x^2$ on $[1, 3]$?

## Question 13 — Challenge

A population is $P(t) = 100 + t^2$. Find its average rate of growth on $[0, 5]$.

# Question Recap & Answer Key

## Question Recap

1. Average rate of $f(x) = x^2 - 4x$ on $[1, 5]$?
2. Average rate of $f(x) = 2x^3$ on $[1, 2]$?
3. Estimate the instantaneous rate of $f(x) = x^2$ at $x = 5$ using $[5, 5.1]$.
4. Average rate of $f(x) = 7x + 1$ on $[2, 5]$?
5. What does the slope of a tangent line measure?
6. Average rate of $f(x) = x^2$ on $[2, 8]$?
7. Average rate of $f(x) = x^2 + 1$ on $[0, 3]$?
8. Average rate of $f(x) = 3x$ on $[2, 6]$?
9. Estimate the instantaneous rate of $f(x) = x^2$ at $x = 6$ using $[6, 6.1]$.
10. Average rate of $f(x) = x^3$ on $[0, 2]$?
11. Does a line have a changing instantaneous rate?
12. Average rate of $f(x) = -x^2$ on $[1, 3]$?
13. A population is $P(t) = 100 + t^2$. Find its average rate of growth on $[0, 5]$.

## ✓ Answer Key

1. 2
2. 14
3. $10.1 \approx 10$
4. 7
5. instantaneous rate of change
6. 10
7. 3
8. 3
9. $12.1 \approx 12$
10. 4
11. no (constant)
12. $-4$
13. $\dfrac{125 - 100}{5} = 5$